Tensor Categories
Proposition 4.2.10: From Left Duals to Rigidity

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Private reader of the 2015 AMS author final copy

Fixed source and scope. *Tensor Categories*, Mathematical Surveys and Monographs 205, American Mathematical Society, 2015, author final copy.
`https://math.mit.edu/~etingof/egnobookfinal.pdf`.
The retained PDF has 362 physical pages; its metadata record creation on May 13, 2015 and modification on July 29, 2015. The complete primary is on physical pages 84–85, printed pages 67–68. Lemma 4.2.11 and the paragraph after its proof belong to this one primary proof.

The chapter opens on physical page 82 (printed 65) with "Let k be an algebraically closed field." The hypothesis here is a *finite ring category with left duals*; right duals are the conclusion. No characteristic-zero, semisimplicity, symmetry, or identification of the double left dual with the identity is added. Selected earlier and forward same-book inputs are printed below as uncounted support. In particular, Proposition 3.3.13(1) is an explicit strong input. Proposition 4.5.7, whose hypothesis already says tensor categories, is not used in this reader to obtain the missing right duals.

Source notice. The source footer states: "Author's final version made available with permission of the publisher, American Mathematical Society. License or copyright restrictions may apply to redistribution; see `http://www.ams.org/publications/ebooks/terms`." This reader remains private. The separate February 14, 2026 corrections are identified at the end; they do not silently replace the 2015 text.

Terminology and explicit prior inputs (uncounted)

Finite abelian categories. Definition 1.8.5 (physical 26, printed 9) says that a k-linear abelian category $\mathcal{C}$ is *finite* if it is equivalent to the category A-mod of finite dimensional modules over a finite dimensional k-algebra A. Definition 1.8.6 gives the equivalent intrinsic conditions:

1. $\mathcal{C}$ has finite dimensional spaces of morphisms;
2. every object of $\mathcal{C}$ has finite length;
3. $\mathcal{C}$ has enough projectives, i.e., every simple object of $\mathcal{C}$ has a projective cover; and
4. there are finitely many isomorphism classes of simple objects.

The first two conditions define *locally finite* (Definition 1.8.1). The notation $\mathcal{O}(\mathcal{C})$ denotes the set of isomorphism classes of simple objects. Length means Jordan–Hölder length. These definitions do not identify the monoidal product with the standard tensor product of modules or supply a fiber functor.

Definition 1.6.5 (physical 23, printed 6) calls P projective if $\mathrm{Hom}_{\mathcal{C}}(P,-)$ is exact. Under the standing finite-length condition, Definition 1.6.6 calls an epimorphism $p : P(X) \to X$ a *projective cover* if $P(X)$ is projective and, whenever $g : P \to X$ is an epimorphism from a projective object, there is an epimorphism $h : P \to P(X)$ with $ph = g$. If a projective cover exists, it is unique up to a non-unique isomorphism. The discussion on physical 27 (printed 10) constructs a projective generator from the covers of the finitely many simple objects. That discussion is background, not a new target.

Ring and tensor categories. Definition 4.2.3 (physical 83, printed 66) reads: A *multiring category* over k is a locally finite k-linear abelian monoidal category $\mathcal{C}$ with bilinear and biexact tensor product. If in addition $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) = k$, we will call $\mathcal{C}$ a *ring category*.

Definition 4.1.1 (physical 82, printed 65) starts with a locally finite k-linear abelian *rigid* monoidal category and calls it a multitensor category when the tensor product is bilinear on morphisms; with $\mathrm{End}_{\mathcal{C}}(\mathbf{1}) \cong k$ it is a tensor category. Definition 2.10.11 (physical 59, printed 42) calls an object rigid if it has left and right duals, and a category rigid if every object is rigid.

Orientations of duals (Definitions 2.10.1–2.10.2). On physical 57 (printed 40), a left dual X^* has evaluation and coevaluation

$$\mathrm{ev}_X : X^* \otimes X \longrightarrow \mathbf{1}, \qquad \mathrm{coev}_X : \mathbf{1} \longrightarrow X \otimes X^*,$$

and a right dual ${}^*X$ has

$$\mathrm{ev}'_X : X \otimes {}^*X \longrightarrow \mathbf{1}, \qquad \mathrm{coev}'_X : \mathbf{1} \longrightarrow {}^*X \otimes X.$$

Their two triangle identities are the definition's (2.43)–(2.46). The preceding coherence convention (Remark 2.9.3) permits suppressing brackets and the associated canonical associativity and unit maps when there is no confusion. These definition excerpts specify the orientations; the four triangle composites and the dual-morphism formulas (2.47)–(2.48) are background and are not reprinted here.

Exercise 2.10.7(b) (physical 59, printed 42) states: if U, V have left (respectively, right) duals, then $U \otimes V$ has left dual $V^* \otimes U^*$ (respectively, right dual ${}^*V \otimes {}^*U$). The exercise's verification is a stated prior input; no new solution is inserted here.

Proposition 2.10.8(a), with its printed proof. Let $\mathcal{C}$ be a monoidal category and let V be an object in $\mathcal{C}$. If V has a left dual V^* then there are natural adjunction isomorphisms

$$\mathrm{Hom}_{\mathcal{C}}(U \otimes V, W) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(U, W \otimes V^*), \tag{2.49}$$

$$\mathrm{Hom}_{\mathcal{C}}(V^* \otimes U, W) \xrightarrow{\sim} \mathrm{Hom}_{\mathcal{C}}(U, V \otimes W). \tag{2.50}$$

Printed proof (physical 59, printed 42). An isomorphism in (2.49) is given by $f \mapsto (f \otimes \mathrm{id}_{V^*}) \circ (\mathrm{id}_U \otimes \mathrm{coev}_V)$ and has the inverse $g \mapsto (\mathrm{id}_W \otimes \mathrm{ev}_V) \circ (g \otimes \mathrm{id}_V)$. Other isomorphisms are similar and are left to the reader as an exercise. □

Only the left-dual part (a) of the statement is selected. The printed proof paragraph is retained in full, including its instruction concerning the other isomorphisms; the right-dual part (b) is not assumed.

Exactness support: Proposition 4.2.9

Physical 84, printed 67; complete statement and proof. If $\mathcal{C}$ is a multiring category with left duals, then the left dualization functor is exact. The same applies to right duals.

Proof. Let $0 \to X \to Y \to Z \to 0$ be an exact sequence. We need to show that the sequence $0 \to Z^* \to Y^* \to X^* \to 0$ is exact. Let T be any object of $\mathcal{C}$, and consider the sequence

$$0 \to \mathrm{Hom}_{\mathcal{C}}(T, Z^*) \to \mathrm{Hom}_{\mathcal{C}}(T, Y^*) \to \mathrm{Hom}_{\mathcal{C}}(T, X^*).$$

By Proposition 2.10.8, it can be written as

$$0 \to \mathrm{Hom}_{\mathcal{C}}(T \otimes Z, \mathbf{1}) \to \mathrm{Hom}_{\mathcal{C}}(T \otimes Y, \mathbf{1}) \to \mathrm{Hom}_{\mathcal{C}}(T \otimes X, \mathbf{1}).$$

Using the definition of a multiring category, we see that the last sequence is exact. This implies that the sequence $0 \to Z^* \to Y^* \to X^*$ is exact.

Similarly, consider the sequence

$$0 \to \mathrm{Hom}_{\mathcal{C}}(X^*, T) \to \mathrm{Hom}_{\mathcal{C}}(Y^*, T) \to \mathrm{Hom}_{\mathcal{C}}(Z^*, T).$$

By Proposition 2.10.8, it can be written as

$$0 \to \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, X \otimes T) \to \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Y \otimes T) \to \mathrm{Hom}_{\mathcal{C}}(\mathbf{1}, Z \otimes T),$$

which is exact since $0 \to X \otimes T \to Y \otimes T \to Z \otimes T$ is exact by the definition of a multiring category. This implies that the sequence $Z^* \to Y^* \to X^* \to 0$ is exact. □

Proposition 4.2.10. *A finite ring category $\mathcal{C}$ with left duals is a tensor category (i.e., it also has right duals).*

Proof. We know that the left duality functor $X \mapsto X^*$ is fully faithful (see Proposition 2.10.8), exact (see Proposition 4.2.9), and monoidal (see Exercise 2.10.7(b)). In particular, this functor respects the Frobenius–Perron dimension by Proposition 3.3.13(1). We claim that this functor sends simple objects to simple ones. Let $d_1 < d_2 < \cdots < d_r$ be the list of possible Frobenius–Perron dimensions of simple objects in $\mathcal{C}$. Let $\mathcal{O}_i(\mathcal{C}) \subset \mathcal{O}(\mathcal{C})$ be the isomorphism classes of simple objects L with $\mathrm{FPdim}(L) = d_i$, $i = 1, \ldots, r$.

Lemma 4.2.11. *Let $L \in \mathcal{O}_i(\mathcal{C})$. Then L^* is simple and there exists $L' \in \mathcal{O}_i(\mathcal{C})$ such that $L \simeq (L')^*$.*

Proof. We prove the lemma by induction in i. The base ($i = 1$) is obvious. If L^* is not simple then there exists a nonzero map $K \to L^*$, where K is simple with $\mathrm{FPdim}(K) < \mathrm{FPdim}(L^*) = d_i$. By the induction assumption $K \simeq (K')^*$, where K' is simple with $\mathrm{FPdim}(K') < d_i$. Since the duality functor is full, we get a nonzero map $L \to K'$, which is a contradiction. Thus we have a map $L \mapsto L^*$ from $\mathcal{O}_i(\mathcal{C})$ to itself. Again by the fullness of duality we see that this map is injective. Since the set $\mathcal{O}_i(\mathcal{C})$ is finite, it is also surjective, which proves the lemma. □

By Lemma 4.2.11 the functor $X \mapsto X^{**}$ induces a permutation of $\mathcal{O}(\mathcal{C})$. Thus there exists a finite power $F = (**)^n$ of the double duality functor such that $F(L) \simeq L$ for any $L \in \mathcal{O}(\mathcal{C})$. Also, the functor F is exact and fully faithful. Let P be the projective cover of $L \in \mathcal{O}(\mathcal{C})$. Since the functor F is full we see that the object $F(P)$ has a unique simple quotient $F(L) \simeq L$. It follows that we have a surjection $P \to F(P)$. Since the length of $F(P)$ equals the length of P, this must be an isomorphism $P \simeq F(P)$. Thus F is surjective on isomorphism classes of projective objects. Since any object of $\mathcal{C}$ is isomorphic to a cokernel of a map of projective objects and the functor F is fully faithful, we get that F is surjective on isomorphism classes of objects and hence is an equivalence. Thus the duality functor $X \mapsto X^*$ is surjective on isomorphism classes of objects. The proposition is proved. □

Grothendieck ring and FPdim inputs (uncounted)

Multiplication and Lemma 4.5.1. *Physical 88–89, printed 71–72.* In Section 1.5 the book introduced the Grothendieck group $\mathrm{Gr}(\mathcal{C})$ of a locally finite abelian category. Now let $\mathcal{C}$ be a multiring category over k. Let X_i, $i \in I$, be representatives of the isomorphism classes of simple objects in $\mathcal{C}$. The tensor product on $\mathcal{C}$ induces a natural multiplication on $\mathrm{Gr}(\mathcal{C})$ defined by

$$X_i X_j := [X_i \otimes X_j] = \sum_{k \in I} [X_i \otimes X_j : X_k] X_k, \qquad i, j \in I. \tag{4.7}$$

Here $[X : Y]$ is the multiplicity of the simple object Y in the Jordan–Hölder series of X.
Lemma 4.5.1. *The above multiplication on* $\mathrm{Gr}(\mathcal{C})$ *is associative.*

Proof. Since the tensor product functor is exact,

$$[(X_i \otimes X_j) \otimes X_p : X_l] = \sum_k [X_i \otimes X_j : X_k][X_k \otimes X_p : X_l], \quad \forall i, j, p, l.$$

On the other hand,

$$[X_i \otimes (X_j \otimes X_p) : X_l] = \sum_k [X_j \otimes X_p : X_k][X_i \otimes X_k : X_l].$$

Therefore, the associativity of the multiplication follows from the isomorphism $(X_i \otimes X_j) \otimes X_p \cong X_i \otimes (X_j \otimes X_p)$. □

Thus, $\mathrm{Gr}(\mathcal{C})$ is a $\mathbb{Z}_+$-ring with the unit $[\mathbf{1}]$. Definition 4.5.2 calls it the Grothendieck ring of $\mathcal{C}$.

Theorem 4.3.8(i): explicit statement input. *Physical 86, printed 69.* In a ring category with left duals, the unit object $\mathbf{1}$ is simple. Its complete printed proof is on physical 86–87 (printed 69–70); that proof is available in the fixed source but is not reprinted here. It is an explicit strong support statement for the following printed proof, not a missing source or an assumption of right duals.

Proposition 4.5.4, complete statement and proof. If $\mathcal{C}$ is a ring category with left duals then $\mathrm{Gr}(\mathcal{C})$ is a transitive unital $\mathbb{Z}_+$-ring in the sense of Definition 3.3.1.

Proof. Recall from Theorem 4.3.8 that the unit object $\mathbf{1}$ in $\mathcal{C}$ is simple. So $\mathrm{Gr}(\mathcal{C})$ is unital. This implies that for any simple objects X, Z of $\mathcal{C}$, the object $X \otimes X^* \otimes Z$ contains Z as a composition factor (as $X \otimes X^*$ contains $\mathbf{1}$ as a composition factor), so one can find a simple object Y_1 occurring in $X^* \otimes Z$ such that Z occurs in $X \otimes Y_1$. Similarly, the object $Z \otimes X^* \otimes X$ contains Z as a composition factor, so one can find a simple object Y_2 occurring in $Z \otimes X^*$ such that Z occurs in $Y_2 \otimes X$. Thus $\mathrm{Gr}(\mathcal{C})$ is transitive. □

In particular, Proposition 4.5.4 shows that for a ring category $\mathcal{C}$ with left duals, we can talk about Frobenius–Perron dimensions of objects and regular objects, see Section 3.3.

Definitions 3.3.1 and 3.3.3 and prior character input. *Physical 70, printed 53.* For a $\mathbb{Z}_+$-ring A with its distinguished basis I, Definition 3.3.1 calls A transitive if for any $X, Z \in I$ there exist $Y_1, Y_2 \in I$ such that XY_1 and Y_2X contain Z with a nonzero coefficient. For a transitive unital $\mathbb{Z}_+$-ring of finite rank, $\mathrm{FPdim}(X)$, $X \in I$, is the maximal non-negative eigenvalue of left multiplication by X, and FPdim is extended from I by additivity. Definition 3.3.3 calls this function the Frobenius–Perron dimension.

Proposition 3.3.6(1),(3) states that $\mathrm{FPdim} : A \to \mathbb{C}$ is a ring homomorphism and is the unique character of A taking non-negative values on I, and that these values are actually strictly positive. These are explicit strong prior inputs; their proof on physical 70–71 (printed 53–54), and the preceding Frobenius–Perron theorem, are not reprinted here.

Proposition 3.3.13(1), with its printed proof. *Physical 72, printed 55.* Let A_1, A_2 be transitive unital $\mathbb{Z}_+$-rings of finite rank with $\mathbb{Z}_+$-bases I_1, I_2. Let $f : A_1 \to A_2$ be a unital homomorphism, whose matrix in the bases I_1, I_2 has non-negative entries. Then f preserves Frobenius–Perron dimensions.

Printed proof of part (1). The function $X \mapsto \mathrm{FPdim}(f(X))$ is a character of A_1 with non-negative values on the basis. By Proposition 3.3.6(3), $\mathrm{FPdim}(f(X)) = \mathrm{FPdim}(X)$ for all X in I. □

Only part (1) and its own proof are selected. The source's final I is retained; the statement specifies I_1, I_2. Parts (2),(3), concerning regular elements, are not needed or reprinted.

Separate 2026 corrections and attribution

The six-page author correction document is dated February 14, 2026: `https://math.mit.edu/~etingof/tensbookcor.pdf`. Its physical page 3 says: "Paragraph before Example 4.5.5: the ring category should be assumed finite." The 2015 paragraph immediately after Proposition 4.5.4 is retained above; this correction is recorded separately. The present primary already assumes finiteness. The same correction page changes the reference in Proposition 4.2.8 and interchanges the words "monomorphisms" and "epimorphisms" in Corollary 4.3.9. Neither of those passages is used as a printed support statement in this reader; they are not silently imported into the primary or Proposition 4.2.9.

The exposition and proof collected here are by the four book authors. The bibliographical notes on physical 105 (printed 88), Section 4.2, specifically credit Proposition 4.2.1 to Deligne–Milne and Proposition 4.2.12 to Kazhdan–Lusztig and related sources. They do not give a separate route byline for Proposition 4.2.10. Those neighbouring credits are not reassigned to this proof. The original base-case, finite-stratum, projective-cover, length and cokernel compressions remain at the authors' printed level; no new proof is added.